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PROJECT SHOWCASE SERIES

KisanRoute

Where Agriculture Meets Intelligence.

An AI-powered platform that understands farmers' questions, classifies them using NLP, and routes them to the right agricultural specialist.



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KisanRoute

The Problem



The Challenge

Farmers ask a wide range of questions — from crop diseases and weather to government schemes and market prices — but these questions are often **unstructured** and **hard to route** to the right expert quickly.

What Problem Are We Solving?

The lack of an intelligent, automatic way to understand a farmer's question and connect it to the **right agricultural expertise.**

Why Is Query Classification Important?

- Understands the intent behind a farmer's query.
- Organizes questions into meaningful agricultural domains.
- Enables faster, more accurate and useful responses.

Why Do Different Questions Need Specialists?

Plant disease	→	Plant Pathology Expert
Weather & alerts	→	Agro-Meteorology Expert
Government schemes	→	Scheme & Credit Expert
Nutrient & soil health	→	Agronomist / Soil Expert
Market prices	→	Market & Mandi Analyst

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KisanRoute

Our Solution

What is KisanRoute?

An AI-powered platform that understands farmers' questions, classifies them using **NLP,** and routes them to the right **agricultural specialist.**



How Does It Help?

- Understands the intent behind a farmer's query.
- Classifies it into the right knowledge domain.
- Recommends the most relevant specialist with confidence.

The 5 Agricultural Knowledge Categories

- Plant Protection
- Weather
- Government Schemes
- Nutrient Management
- Market Information

What Makes It Different?

- Real NLP Classification
- Trained & Evaluated ML Model
- Confidence-Based Routing
- Interactive Insights & Analysis
- Privacy-First & Lightweight

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KisanRoute

Technology Behind the Project

Frontend

Next.js 15
Modern, fast and responsive UI

TypeScript
Type-safe and scalable development

Backend

Python
Core logic and ML integration

FastAPI
High-performance REST APIs

NLP & Feature Engineering

NLP
Processing and understanding farmer queries

TF-IDF
Converts text into meaningful numerical features

Machine Learning

Logistic Regression
Strong, reliable baseline model

Naive Bayes
Final model (91.43% Accuracy, 91.25% Macro-F1)

The KisanRoute Pipeline

Farmer Query
(Natural Language)
→
Text Preprocessing
(Cleaning & Normalization)
→
TF-IDF Vectorization
(N-gram Features)
→
ML Classification
(Logistic Regression / Naive Bayes)
→
Confidence Scoring
(Calibrated Prediction)
→
Specialist Routing
(Right Domain & Expert)

Intelligent Specialist Routing

Automatically maps each query to the most relevant agricultural knowledge domain and recommends the **right specialist** with confidence.

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KisanRoute

Project Demo

Live Website https://kisanroute.vercel.app/



In This Demo, I'll Show:

- The Home experience and key idea
- How the solution works (NLP pipeline)
- The Intelligence Lab (live query classification)
- Model Insights and performance
- The 5 Knowledge Disciplines and specialist routing

Let's Explore the Live Website!